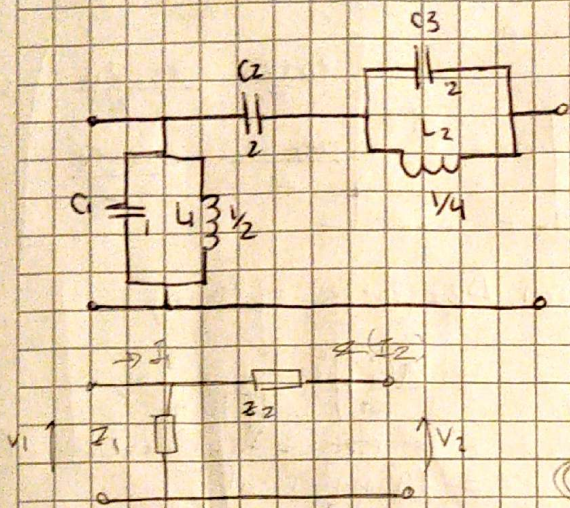


TP 4 CUADRIPOLOS

1 Obtener Z, Y, T de



es como una - es como una - es como una

$$Z = Z_1 // Z_2 \quad \rightarrow \quad Z_1 = Z_{C1} // Z_{L1} = \frac{1}{\frac{1}{s} + sL_1} = \frac{sL_1}{1 + s^2 L_1 C_1}$$

$$Z_2 = Z_{C2} + Z_{L2} // Z_{C3} = \frac{1}{s} + \frac{sL_2}{1 + s^2 L_2 C_3} = \frac{1 + s^2 L_2 C_3 + s^2 L_2 C_2}{s(1 + s^2 L_2 C_3)}$$

$$Z = \frac{sL_1}{1 + s^2 L_1 C_1} \cdot \frac{s(1 + s^2 L_2 C_3)}{1 + s^2 L_2 C_3 + s^2 L_2 C_2} = \frac{s^2 L_1 (1 + s^2 L_2 C_3)}{(1 + s^2 L_1 C_1)(1 + s^2 L_2 C_3 + s^2 L_2 C_2)}$$

$Z_1 \rightarrow Y$

$Z_2 \rightarrow Z$

$A = \frac{V_1}{V_2} \Big|_{I_2=0} = 1 \quad B = \frac{V_1}{-I_2} \Big|_{V_2=0} = Z \quad C = \frac{I_1}{V_2} \Big|_{I_2=0} = Y \quad D = \frac{-I_1}{-I_2} \Big|_{V_2=0} = 1 + ZY$

$$T = \begin{bmatrix} 1 & Z \\ Y & 1 + YZ \end{bmatrix}$$

$$Z = \begin{bmatrix} \frac{A}{C} & \frac{1}{C} \\ \frac{AD-B}{C} & \frac{D}{C} \end{bmatrix}$$

$\Delta = \frac{Z_{11}}{Z_{21}} \quad B = \frac{\Delta Z}{Z_{21}} \quad C = \frac{1}{Z_{21}} \quad D = \frac{-Z_{22}}{Z_{21}}$

$Z_{11} = \frac{A}{C} \quad B = \frac{\Delta Z}{Z_{21}} \quad Z_{21} = \frac{1}{C} \quad Z_{22} = D \cdot Z_{21} = \frac{D}{C}$

$\frac{B}{C} = \frac{A}{C} \cdot \frac{D}{C} = \frac{1}{C^2} Z_{12}$

$Z_{12} = \frac{AD-B}{C} \Rightarrow Z_{12} = \frac{AD-B}{C}$

$Y = Z^{-1} \quad \Delta Z = \frac{A}{C} \cdot \frac{D}{C} - \frac{1}{C} \cdot \frac{(AD-B)}{C}$

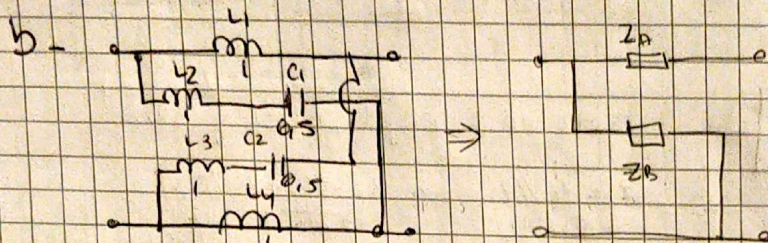
$= \frac{AD}{C^2} - \frac{AD}{C^2} + \frac{B}{C} = \frac{B}{C}$

$$Y = \begin{bmatrix} \frac{D}{B} & \frac{(AD-BC)}{B} \\ \frac{1}{B} & \frac{A}{B} \end{bmatrix}$$

$y_{11} = \frac{Z_{22}}{\Delta Z} = \frac{D}{B} \quad y_{12} = -\frac{Z_{12}}{\Delta Z} = -\frac{(AD-BC)}{B} = \frac{BC-AD}{B}$

$y_{21} = -\frac{Z_{21}}{\Delta Z} = \frac{1}{B} \quad y_{22} = \frac{Z_{11}}{\Delta Z} = \frac{A}{B}$

$$T = \begin{bmatrix} 1 & \frac{s^2+1}{s(s^2+2)} \\ \frac{s^2+2}{s} & \frac{2s^2+1}{s^2} \end{bmatrix} \quad Z = \begin{bmatrix} \frac{s}{s^2+2} & \frac{s}{s^2+2} \\ \frac{s}{s^2+2} & \frac{2s^2+1}{s(s^2+2)} \end{bmatrix} \quad Y = \begin{bmatrix} \frac{(s^2+2)(2s^2+1)}{s(s^2+1)} & -\frac{s(s^2+2)}{s^2+1} \\ -\frac{s(s^2+2)}{s^2+1} & \frac{s(s^2+2)}{s^2+1} \end{bmatrix}$$



$$\begin{aligned} \cdot Z_A = sL_1 &= s \\ \cdot Z_B = sL_2 + \frac{1}{sC_1} &= \frac{s^2 L_2 C_1 + 1}{sC_1} \\ &= \frac{s^2 \cdot \frac{1}{2} + 1}{s \cdot \frac{1}{2}} = \frac{s^2 + 2}{s} \\ \cdot Z_C = Z_D &= s^2 + 2 \\ \cdot Z_D = Z_A &= s \end{aligned}$$

is symmetric

$$\begin{aligned} \text{con } \Delta Z &= \left(\frac{Z_A + Z_B}{2} \right)^2 - \left(\frac{Z_A Z_B}{2} \right)^2 \\ &= \left(\frac{s^2 + 1}{s} \right)^2 - \left(\frac{1}{s} \right)^2 \\ &= \frac{(s^2 + 1)^2 - 1}{s^2} \\ &= \frac{4s^2 + 2s^2}{s^2} = 6 \end{aligned}$$

$$A = \frac{Z_{11}}{Z_{22}} = \frac{\frac{s^2+1}{s}}{\frac{s}{s^2+2}} = \frac{s^2+1}{s} \cdot \frac{s^2+2}{s} = \frac{(s^2+1)(s^2+2)}{s^2}$$

$$B = \Delta Z = \frac{\Delta Z}{Z_{21}} = \frac{\Delta Z}{s}$$

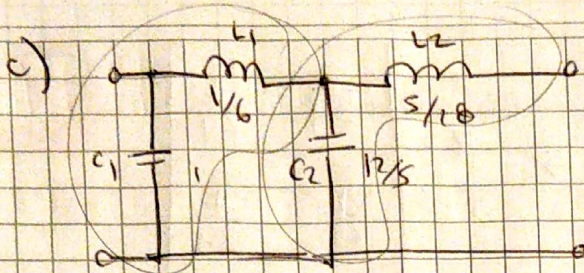
$$C = \frac{1}{Z_{21}} = \frac{1}{\frac{s}{s^2+2}} = \frac{s^2+2}{s}$$

$$D = \frac{Z_{22}}{Z_{21}} = \frac{\frac{s^2+1}{s}}{\frac{s}{s^2+2}} = \frac{(s^2+1)(s^2+2)}{s^2}$$

$$\begin{aligned} \cdot y_{11} &= \frac{-Z_{22}}{\Delta Z} = \frac{-\frac{s^2+1}{s}}{\frac{6}{s^2}} = -\frac{(s^2+1)s}{6} \\ \cdot y_{12} &= \frac{-Z_{21}}{\Delta Z} = \frac{-\frac{s}{s^2+2}}{\frac{6}{s^2}} = -\frac{s^3}{6(s^2+2)} \\ \cdot y_{21} &= \frac{-Z_{11}}{\Delta Z} = \frac{-\frac{s^2+1}{s}}{\frac{6}{s^2}} = -\frac{(s^2+1)s}{6} \\ \cdot y_{22} &= \frac{-Z_{12}}{\Delta Z} = \frac{-\frac{s}{s^2+2}}{\frac{6}{s^2}} = -\frac{s^3}{6(s^2+2)} \end{aligned}$$

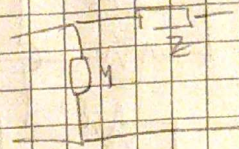
$$T = \begin{bmatrix} (s^2+1) & \Delta Z s \\ s & (s^2+1) \end{bmatrix} \quad Z = \begin{bmatrix} \frac{s^2+1}{s} & \frac{1}{s} \\ \frac{1}{s} & \frac{s^2+1}{s} \end{bmatrix} \quad Y = \begin{bmatrix} \frac{s^2+1}{\Delta Z s} & -\frac{1}{\Delta Z s} \\ -\frac{1}{\Delta Z s} & \frac{s^2+1}{\Delta Z s} \end{bmatrix}$$

$$Y_2 = \frac{1}{C} \cdot \frac{1}{sL}$$



como el a) pero en cascada

$$T_L = \begin{bmatrix} 1 & Z \\ Y & 1+YZ \end{bmatrix}$$



$$T_{total} = T_1 \cdot T_2 = \begin{bmatrix} 1 & Z_1 \\ Y_1 & 1+YZ_1 \end{bmatrix} \begin{bmatrix} 1 & Z_2 \\ Y_2 & 1+YZ_2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{5}{6} \\ \frac{1}{5} & 1+\frac{5}{6} \end{bmatrix} \cdot \begin{bmatrix} 1 & \frac{5}{18} \\ \frac{12}{5} & 1+\frac{5}{18} \end{bmatrix}$$

$$= \begin{bmatrix} 1+\frac{5}{6} \cdot \frac{12}{5} & \frac{5}{6} + \frac{5}{6} \cdot \frac{5}{18} \\ \frac{1}{5} + \frac{12}{5} \cdot \frac{5}{6} & \frac{1}{5} + \frac{12}{5} \cdot \frac{5}{6} \cdot \frac{5}{18} \end{bmatrix} = \begin{bmatrix} \frac{5+25^2}{5} & \frac{25^3+85}{18} \\ \frac{205+725+125^2}{30} & \frac{25^4+205^2+18}{18} \end{bmatrix}$$

$$\Rightarrow \Delta T = 1$$

usando relaciones de ondas

$$y_{11} = \frac{0}{6}, y_{12} = -\frac{\Delta T}{6}, y_{21} = -\frac{1}{6}, y_{22} = \frac{\Delta}{6}, z_{11} = \frac{\Delta}{C}, z_{12} = \frac{1}{C}, z_{21} = \frac{\Delta T}{C}, z_{22} = \frac{0}{C}$$

$$Y = \begin{bmatrix} \frac{25^4+205^2+18}{18} & -\frac{18}{25^3+85} \\ \frac{25^3+85}{18} & -\frac{18}{25^3+85} \end{bmatrix} = \begin{bmatrix} \frac{25^4+205^2+18}{25^3+85} & -\frac{18}{25^3+85} \\ -\frac{18}{25^3+85} & \frac{15+25^2+18}{25^3+85} \end{bmatrix}$$

$$Z = \begin{bmatrix} \frac{5+25^2}{5} & \frac{30}{205+725+125^2} \\ \frac{305+725+125^2}{30} & \frac{25^4+205^2+18}{18} \end{bmatrix} = \begin{bmatrix} \frac{30(5+25^2)}{5(125^2+1025)} & \frac{30}{125^2+1025} \\ \frac{30}{125^2+1025} & \frac{30(25^4+205^2+18)}{18(125^2+1025)} \end{bmatrix}$$